Applebaum's stable trace-asymptotic conjecture was answered by arXiv:1308.4944

Literature-status note for `fa_banach_001`

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Status. `literature_already_answered (full conjecture)`. No new mathematical result is claimed by this packet.

Original conjecture

Let G be a compact semisimple Lie group of dimension d. For the central α-stable semigroup constructed from the Casimir operator, Applebaum uses the exponent $\eta(u) = |u|^\alpha$ and denotes its continuous density by k_t. In Section 6, arXiv PDF page 21 (journal page 2494), he conjectures that, for $0 < \alpha \le 2$,

$$k_t(e) \sim Ct^{-d/\alpha} \qquad (t \downarrow 0), \tag{1}$$

with $C > 0$; the Cauchy conjecture is the case $\alpha = 1$. The same section has already established $k_t(e) = \mathrm{Tr}(T_t)$.

Explicit later answer

Banuelos and Baudoin state in the abstract and first paragraph of arXiv:1308.4944, PDF page 1, that they address Applebaum's conjecture and that their purpose is to show it is indeed true [2]. They cite Applebaum's paper as reference [1] and identify the conjecture at its journal pages 2493–2494.

Their pages 2–3 take

$$\psi(\lambda) = \lambda^{\alpha/2}, \qquad 0 < \alpha < 2,$$

so that the subordinate generator is $(-\Delta)^{\alpha/2}$. If M is a compact Riemannian n-manifold, Weyl's law and a Tauberian argument give

$$\mathrm{Tr}(Q_t) \sim \frac{\mathrm{Vol}(M)\,\Gamma(n/\alpha + 1)}{\Gamma(n/2+1)(4\pi)^{n/2}}\, t^{-n/\alpha}. \tag{2}$$

The unnumbered theorem and equation (9) on PDF page 3 prove a more general regular-variation version. Immediately after (9), they observe that on a compact Lie group with normalized Haar measure the trace asymptotic gives the on-diagonal kernel asymptotic. Applying this to the bi-invariant Laplacian on G yields (1). This is exactly Applebaum's construction, because $\eta(\sqrt{\kappa}) = \kappa^{\alpha/2}$ on every Casimir eigenspace.

The answering note treats $0 < \alpha < 2$. The endpoint $\alpha = 2$ is the classical heat-kernel case, which Applebaum explicitly records as already known in the same section. Consequently the combined literature answers the full stated range $0 < \alpha \le 2$, and even proves the trace result on every closed Riemannian manifold.

Search and scope

The four run indexes, exact-title and DOI hits in the local parsed arXiv corpus, exact-phrase web searches, and the 16 OpenAlex-indexed citing works were checked through 26 August 2026. arXiv:1308.4944 is a direct, explicit answer rather than an agent-identified implication. Later papers by Fahrenwaldt and by Applebaum–Le Ngan corroborate its use. No portion of Applebaum's stable-asymptotic conjecture is claimed open here.

References

[1] D. Applebaum, *Infinitely divisible central probability measures on compact Lie groups—regularity, semigroups and transition kernels*, Ann. Probab. **39** (2011), 2474–2496; arXiv:1006.4711, https://arxiv.org/abs/1006.4711.

[2] R. Banuelos and F. Baudoin, *Trace asymptotics for subordinate semigroups*, arXiv:1308.4944 (2013), https://arxiv.org/abs/1308.4944.